Anna Fung

WORK IS A REAL KILLER

How the economy influences traffic fatalities

Hypothesis

When unemployment rises, fewer people drive to work, leading to fewer traffic fatalities overall

Variable	Description
totfatrte	total traffic fatality rate (per 100,000 population)
unem	state unemployment rate (%)
zerotol	zero tolerance underage DUI law indicator
minage	minimum legal drinking age indicator
bac08	BAC 0.08 law indicator
bac10	BAC 0.10 law indicator
wkndfatrte	weekend traffic fatality rate (per 100,000 population)

*All models have fixed effects for state and robust standard errors for heteroskedasticity

Weekend Fatalities: Model 2

$$wkndfatrte_i = \beta_1 + \beta_2 unem_i + \beta_3 zerotol_i + \beta_4 minage_i + \beta_5 bac08_i + \beta_6 bac10_i + \varepsilon_i$$

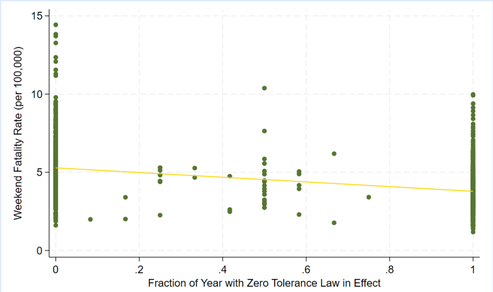
Variable	Coefficient	Robust Standard Errors	T-value
unem	-0.8322	0.0302	-2.75
zerotol	-0.9365	0.0928	-10.09
minage	-0.3529	0.0774	-4.56
bac08	-1.7642	0.3788	-4.66
bac10	-1.2392	0.2727	-4.54
constant	13.9419	1.5387	9.06

- Alcohol-related policies significantly reduce weekend traffic fatalities
- Weekend fatalities are less sensitive to unemployment

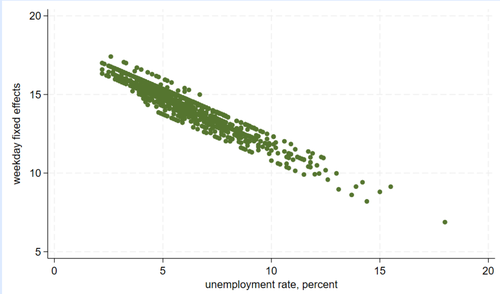
Model 1

$$totfatrte_i = \beta_1 + \beta_2 unem_i + \beta_3 zerotol_i + \beta_4 minage_i + \beta_5 bac08_i + \beta_6 bac10_i + \varepsilon_i$$

Variable	Coefficient	Robust Standard Error	T-value
unem	-0.4522	0.0923	-4.9
zerotol	-2.1616	0.3914	-5.52
minage	-0.7132	0.2376	-3
bac08	-4.7147	0.9979	-4.72
bac10	-3.1615	0.7415	-4.26
constant	40.253	4.9972	8.06



Model 2



Model 3

Conclusion

- Weekday fatalities are strongly influenced by economic conditions
- Weekend fatalities respond more to alcohol polices
- Alcohol policies matters, even with fixed effects and robust errors, policies consistently reduce fatalities
- Economic conditions cannot explain all fatalities

So what does this mean?

- Higher unemployment is linked to fewer total traffic fatalities
- Alcohol related policies show meaningful effects on fatality rates
- The collective policy effect is still significant even with fixed effects and robust errors
- We can't jump to the conclusion that "bad economy=safer roads"
- Weekday and weekend driving patterns differ, what happens when we separate the two?

Weekday Fatalities: Model 3

wkdayfatrte=totfatrte-wkndfatrte

$$wkdayfatrte_i = \beta_1 + \beta_2 unem_i + \beta_3 zerotol_i + \beta_4 minage_i + \beta_5 bac08_i + \beta_6 bac10_i + \varepsilon_i$$

Variable	Coefficient	Robust Standard Errors	T-value
unem	-0.3690	0.0676	-5.46
zerotol	-1.2250	0.3180	-3.85
minage	-0.3603	0.1807	-1.99
bac08	-2.9506	0.6573	-4.49
bac10	-1.9222	0.5141	-3.74
constant	26.3110	3.8552	6.82

- Higher unemployment is associated with fewer weekday traffic fatalities
- Weekday fatalities respond stronger to economic conditions than to alcohol-related policies